

Plane ticket

A travel agency keeps record of the prices of plane tickets and their corresponding distances. Write a program that gives the price and index of the plane ticket to the farthest city.

Input

The first line of the standard input contains the number of cities ($1 \leq N \leq 100$). The next N lines contain the distance of a city from the Budapest airport ($1 \leq D \leq 20000$) and the price of the plane ticket ($1 \leq P \leq 2000000$) each.

Output

The first line of the standard output should contain the index of the plane ticket to the furthest city and the second line should contain the price of that ticket (if there's multiple solution, you should give the first).

Example

Input

```
7
50 30000
1900 200000
2000 150000
900 38000
600 150000
2100 30000
2100 50000
```

Output

```
6
30000
```

specification tool: <https://progalap.elte.hu/specifikacio/>

algorithm tool: <https://progalap.elte.hu/stuki/>